

Ex: Let $f(x) = \begin{cases} \frac{1}{x+2}, & x < -2 \\ x^2 - 5, & -2 \leq x \leq 3 \\ \sqrt{x+13}, & x > 3 \end{cases}$

Limit $x \rightarrow -2^-$, $x < -3/2$

$\lim_{x \rightarrow -2^-} \frac{1}{x+2} = \frac{1}{-2+2} = \frac{1}{0} = \infty$

$x = -2$, $x = 3$

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Find (a) $\lim_{x \rightarrow 0} f(x)$, (b) $\lim_{x \rightarrow -2} f(x)$, (c) $\lim_{x \rightarrow 3} f(x)$

Sol: (b) $\lim_{x \rightarrow -2} f(x)$

L.H.L = $\lim_{x \rightarrow -2^-} f(x) = \lim_{h \rightarrow 0} \frac{1}{-2-h+2} = \lim_{h \rightarrow 0} -\frac{1}{h} = -\infty$

R.H.L = $\lim_{x \rightarrow -2^+} f(x) = \lim_{h \rightarrow 0} (-2+h)^2 - 5 = \lim_{h \rightarrow 0} (-2+h)^2 - 5 = 4 - 5 = -1$

L.H.L $\neq$ R.H.L

$\lim_{x \rightarrow -2} f(x)$ does not exist.

(c) $\lim_{x \rightarrow 3} f(x)$

R.H.L = $\lim_{x \rightarrow 3^+} f(x)$

L.H.L = $\lim_{x \rightarrow 3^-} f(x)$

$= \lim_{h \rightarrow 0} f(3+h)$

$x < 3$, $x = 3$
 $x > 3$, $x = 3$

$= \lim_{h \rightarrow 0} f(3-h)$

$= \lim_{h \rightarrow 0} (3-h)^2 - 5$

$= 9 - 5 = 4$

$= \lim_{h \rightarrow 0} \sqrt{3+h+13}$

$= \sqrt{16}$

$= 4$

L.H.L = R.H.L = 4, $\lim_{x \rightarrow 3} f(x) = 4$. Ans

(a) $\lim_{x \rightarrow 0} f(x) = \lim_{x \rightarrow 0} x^2 - 5 = -5$. Ans

$-\sqrt{16} = -4$, $\sqrt{16} = 4$

$\sqrt{16} \neq \pm 4$

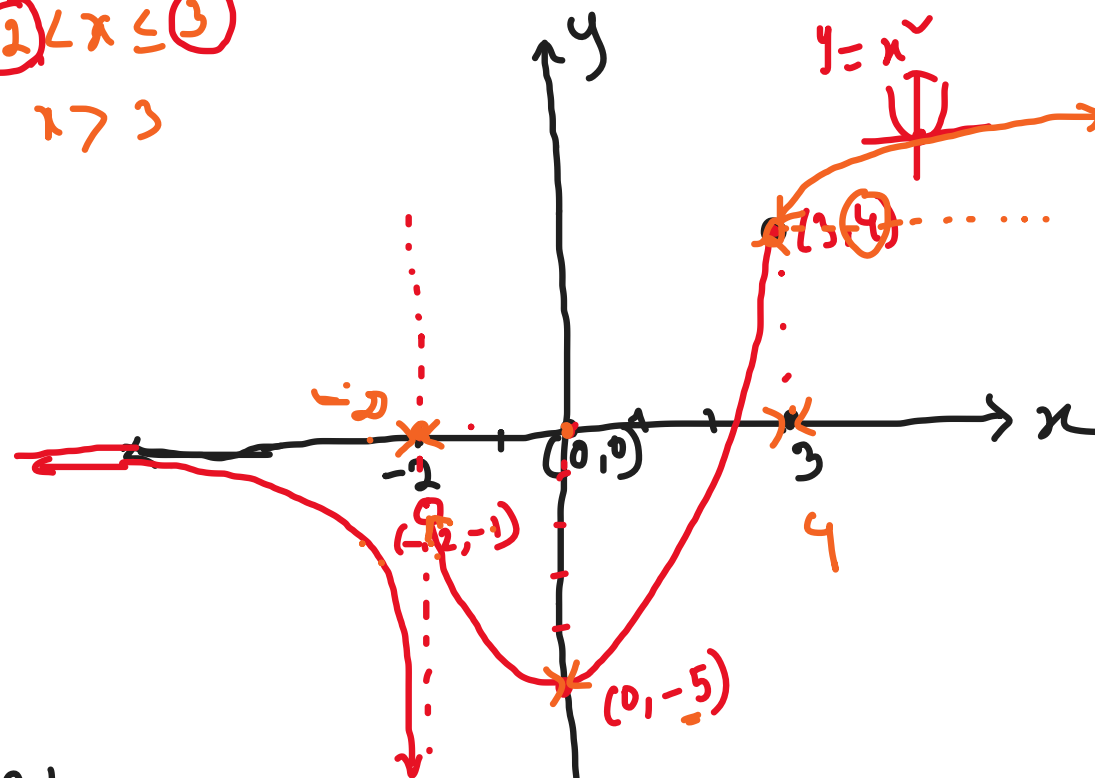
$x^2 = 4$

$y^2 = 4x \Rightarrow y = \pm \sqrt{4x}$

$\Rightarrow x = \pm \sqrt{4} = \pm 2$

$x = 4$
 $y = \pm \sqrt{4} = \pm 2$

$f(x) = \begin{cases} \frac{1}{x+2}, & x < -2 \\ x^2 - 5, & -2 \leq x \leq 3 \\ \sqrt{x+13}, & x > 3 \end{cases}$



$\frac{1}{x} \rightarrow \infty = 0$

$\frac{1}{x+2} \rightarrow \infty = -2$

D: $\mathbb{R} - \{-2\}$
R: $\mathbb{R} - \{0\}$

$y = \sqrt{x}$

Algebraic Manipulation:

Ex: Find $\lim_{x \rightarrow 3} \frac{x^2 - 6x + 9}{x - 3}$

$\frac{9 - 18 + 9}{0} = \frac{0}{0}$ in determinate form

Sol: $\lim_{x \rightarrow 3} \frac{(x-3)^2}{(x-3)}$

$= \lim_{x \rightarrow 3} x - 3 = 3 - 3 = 0$. Ans

Limit at infinity

Ex: Find $\lim_{x \rightarrow \infty} \frac{3x^2 + 5}{6x - 9}$

$\left[\frac{3x^2 + 5}{6x - 9} = \frac{\infty + 5}{\infty - 9} = \frac{\infty}{\infty} \times \text{indeterminate form} \right]$

$= \lim_{x \rightarrow \infty} \frac{x(3 + 5/x)}{x(6 - 9/x)}$

$= \frac{3 + 5/0}{6 - 9/0} = \frac{\infty}{\infty} = \frac{1}{2}$. Ans

$\lim_{x \rightarrow \infty} \frac{5}{x} = 0$